NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	24/12/2025	Issued For Review	ZLN	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	967376,210	10,414	
PM-8593	425000,093	907528,777	19,046	

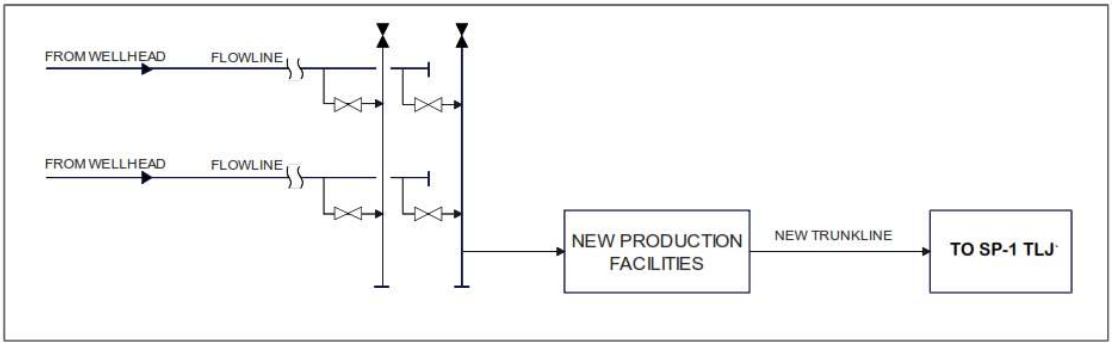
Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan

pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

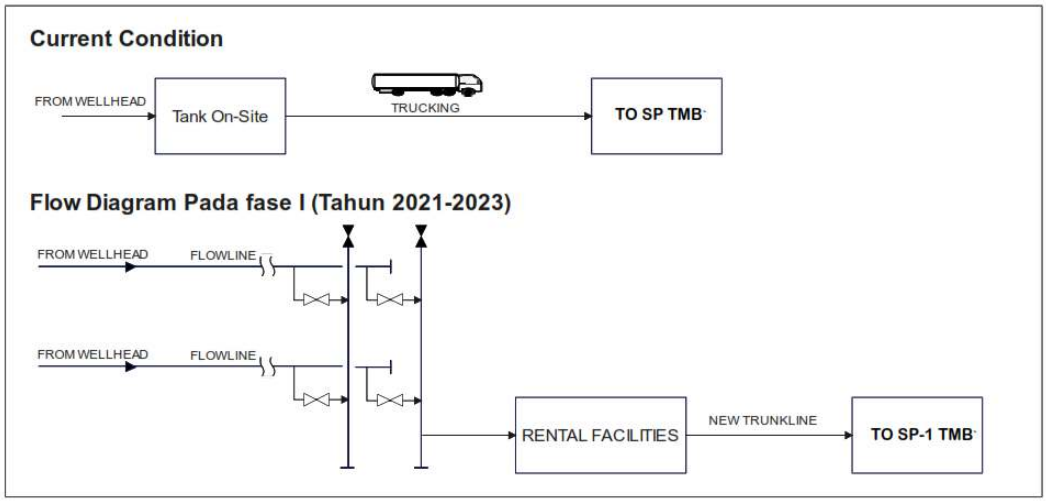
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 2. Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 3. Rencana Pola Operasi Fase

1.2 Tujuan

Dokumen ini menjelaskan dipersiapkan sebagai Mechanical Datasheet for Air Compressor Package (Z-006) & Air Dryer Package (Z-005) sebagai bagian dari desain yang harus diikuti oieh KONTRAKTOR dan VENDOR untuk proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat- Karangan Timur, Field Prabumulih.

1.3 Ruang Lingkup

Mechanical Datasheet ini meliputi ketentuan minimal dalam mengatur desain. material,pabrikasi, pemeriksaan, pengujian, persiapan, pengecatan dan persiapan untuk pengiriman Air Compressor Package (Z-006) & Air Dryer Package (Z-005) untuk dipasang di proyek Pembangunan Fasilitas Stasiun Pengumpul Lapangan Tanjung Miring Barat — Karangan Timur, Field Prabumulih.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik

	terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

DOKUMEN PROYEK

1. TMBP-SP001-ME-DBS-D101 — Mechanical Design Basis
2. TMBP-SP001-PR-PID-D115 — PID Utility and Instrument Air System
3. TMBP-SP001-ME-SPE-0105 — Specification for Air Compressor & Dryer Package

3. Daftar Singkatan

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

BOPD	*Barrel Oil Per Day*
BPD	*Barrel Per Day*
BWPD	*Barrel Water Per Day*
EOR	*Enhanced Oil Recovery*
FEED	*Front End Engineering Design*
KRG	*Karangan*
MMSCFD	*Million Standard Cubic Feet per Day*
NPS	*Nominal Pipe Size*
PB	*Production Battery*

PHR	*Pertamina Hulu Rokan*
PT	*Perseroan Terbatas*
SP	*Stasiun Pengumpul*
SCH	*Schedule*
TMB	*Tanjung Miring Barat*
TMT	*Tanjung Miring Timur*
WK	*Wilayah Kerja*

Applicable to	: **PROPOSAL**	**Client**	: **PT. PERTAMINA EP**
Tag No.	: **Z-006 & Z-005**	**Quantity**	: **2 Units**
Service	: **Air Compressor and Air Dryer Package**	**Code & Standard**	: **API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
P & ID No.	: **TMBP-01-ME-SPE-154**	**Manufacturer**	: **As per PML**

EXTENT OF PACKAGE

1	**SITE CONDITION**							Rev.
2	Altitude (above sea level)		m :	**MSL + 260**	Tropicalization	:	**Yes**	
3	Ambient Temperatur	minimum	°C :	**23.0**	Winterization	:	**No**	
4		maximum	°C :	**34**	Area Classification	:	**Non-Hazardouse**	
5	Seismic Factor		:	**SS : 0.0422 g**	Location	:	**Outdoor (Under Roof), Unheated**	
6			:	**S1 : 0.0627 g**				
7	Relative Humidity	maximum	% :	**88**				
8								
9								

SCOPE OF SUPPLY

VENDOR shall provide Air Compressor Package and Air Dryer Package with the following :

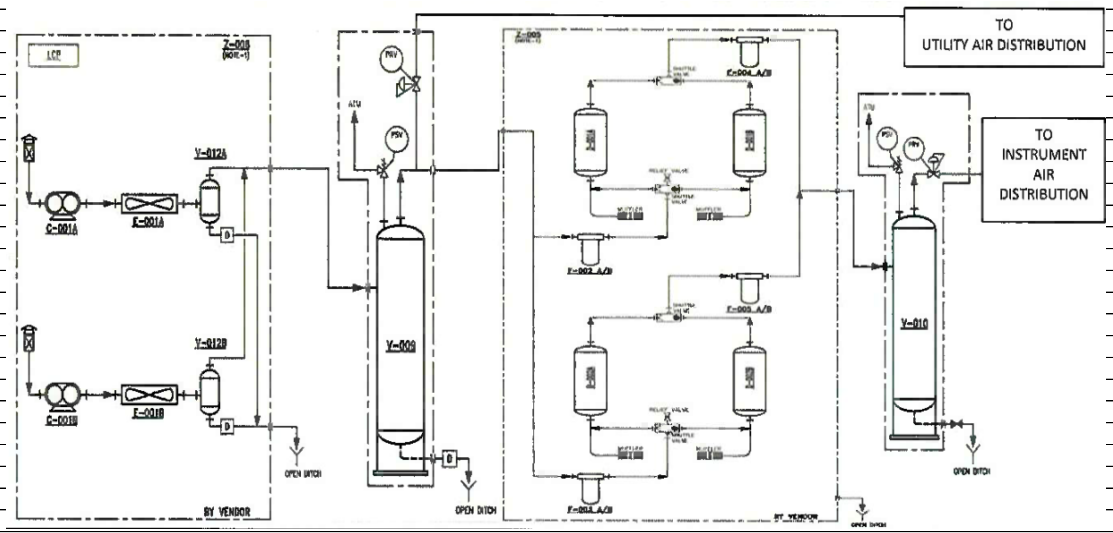
1. Two (2) unit of Air Compressor & Air Dryer Package include but not limited to the following :
   a. 2 (Two) Air Compressor Packages , with configuration 2x100%, which each unit consist of:
      - Air cooled Oil injected screw-type compressor
      - Electric Motor Driver c/w coupling and guard (including space heater f required)
      - Compressor dry type Intake Filter/ Silencer
      - Local Control Panel including motor starter (Soft starter), protection, contactor, circuit breaker, manual start stop push button, indication lamp, etc.
      - Inter / After Cooler
      - Compressor Condensate Separator
      - Discharge pulsation dampers, (if required)
      - Acoustic/ weatherproof enclosure for each compressor package
      - Lead lag controller manager
      - Lube oil system, include:
        - Oil separator
        - Oil cooler
        - Oil Filter
      - Autodrain condensate
   b. Two (2) units Air Dryer (Heatless Type) with Desiccant
   c. Two (2) units Pre Filter per dryer chamber. Total Four (4).
   d. Two (2) units After Filter per dryer chamber. Total Four (4).
2. One (1) Utility Air Receiver
3. One (1) Instrument Air Receiver
4. Instrumentation include Safety Valve, Pressure Indicator, Temperature Indicator, Level Switch, etc.
5. Cabling (power and control wiring) and termination from LCP to motor and within the skid
6. Piping interconnection (including pipe support, valve and fittings) within the package
7. Automatic drain traps with valve
8. Structural skid including drip pan, earthing and lifting lugs
9. Nameplates c/w bracket

APPLICABLE DOCUMENTS

VENDOR to confirm that fabrication, inspection, and testing shall follow the reference Codes, Standard, Project Specifications such mention bellow :

- API 619 Latest Edition	Rotary Type Positive Displacement Compressor for Pertoleum, Petrochemical and Natural Gas Industries
- TMBP-SP001-ME-SPE-0105	Specification for Air Compressor & Dryer Package
- TMBP-SP001-ME-SPE-D101	Specification for Pressure Vessel (ASME)
- TMBP-SP001-PI-SPE-D109	Piping Painting & Lettering Specification
- TMBP-SP001-PI-SPE-D102	Specification for Piping Flexibility Analysis
- TMBP-SP001-EL-DBS-D101	Electrical Design Criteria
- TMBP-SP001-IN-DBS-D101	Instrument and Control System Design Basis
- TMBP-SP001-EL-SPE-D106	Specification for Induction Motor
- TMBP-SP001-IN-SPE-D105	General Specification for Instrument
- TMBP-SP001-PI-SPE-D101	Piping Design Specification
- TMBP-SP001-PI-SPE-D107	Piping Material Specification
- TMBP-01-PR-PID-120	PID Utility and Instrument Air System

SKECTH



Applicable to	:	**PROPOSAL**	**Client**	:	**PT. PERTAMINA EP**
Tag No.	:	**Z-006 & Z-005**	**Quantity**	:	**2 Units**
Service	:	**Air Compressor and Air Dryer Package**	**Code & Standard**	:	**API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
P & ID No.	:	**TMBP-01-ME-SPE-154**	**Manufacturer**	:	**As per PML**

AIR COMPRESSOR (Z-006)

							Rev.
1	**PROCESS DATA**						
2	Suction Pressure	**ATMOSPHERIC**	ATM	Capacity per each Air Compressor	**170**	SCFM	
	Operating Pressure (Min/Nor/Max)	**100 (*) / * / 150 (*)**	psig	Design Pressure	**200 (*)**	psig	
3	Operating Temperature	**(*)**	°F	Design Temperature	**200 (*)**	psig	
4	**MECHANICAL DATA**						
5	**COMPRESSOR**			**DRIVER (Note 4)**			
6	Equipment No.	**Z-006**		Type:	**Induction Motor**		
7	Type **(Note 7)**	**Oil-Injected Screw Comp.**		Specification	**TMBP-SP001-EL-SPE-D106**		
8	No. required	**two (2) units**		Make/Model	*		
9	Make/Model	*		No. required	**Two (2)**		
10	No. of Stages	*		Frame /Enclosure	**TEFC**		
11	Operation Mode	**Intermittent, Lead-Lag**		Speed (rated/indle)	*	rpm	
12	Pressure range (Inlet / Outlet)	*	psig	Nominal Power	*	kW	
13	Capacity per compressor	**170**	SCFM	Phase / Hz / Rated Voltage (Volt)	**3 / 50 / 400**		
14	Design pressure	**200**	psig	Service Factor	**1.0**		
15	Receiver Tank Volume			Winding Insulation Class	**F**		
16	Transmision between compressor & driver	**Direct Driven**		Temperatur Rise Class	**B**		
17	Material :	**Note 3**		Motor Starter Type	**DOL**		
18	Frame	*		Motor Enlcosure	**IP55**		
19	Finger valve	*		Noise Level Requirement @ 3 ft distance	**85 dB**		
20	**COMPRESSOR SEPARATOR**			**AFTER COOLER**			
21	Separator No.			After Cooler No.	**-**		
22	No. Required	**two (2) units**		No. Required	**two (2) units**		
23	Installed at	**Outlet Compressor**		Type	**Air Cooled with Fan**		
24	Pressure Drop	*	psig	Fan Motor Power	*		
25	Filter Type	**Cartridge**		Material	*		
26	Filter Size	**2**	microns	Driver	**Induction Motor**		
27	Efficiency	**95**	%				

28	**INSTRUMENT INFORMATION**					
29	**ALARMS AND SHUTDOWNS**				**CONTROL PANEL (Note 5)**	
30		Indicator	Alarm	Shutdown		
31	Compressor Disch. Pressure	√	√ (H,L)	√ (HH)	Instrument area classification	: **Unclassified**
32	Compressor Disch. Temp.	√	√ (H)	√ (HH)	Control Panel	: **Yes - PLC based**
33	Air receiver tank liquid level	√	√ (H)		Panel Location	: **Local Panel**
34	Common Alarm (UCP)	√			Panel Protection	: **IP 65 (outdoor with shelter)**
35	Trip Status	√			Panel Power Supply	: **400VAC/ 3ph/ 50Hz**
36	After Cooler Disch. Temperature	√	√ (H)	√ (HH)		
	After Cooler Fan Motor Vibration	√	√ (H)	√ (HH)		
37	Hight Temp. Cooler Switch	√	√	√		
38	Motor Overload	√	√	√		

39	**CONNECTIONS (Note 2)**						**WEIGHTS , DIMENSIONS, & SIZE**		
40		Qty	Size	Rating	Facing	Remarks	Working Weight	: *	lbs
41			(in)				Dry Weight per unit	: *	lbs
42	Discharge Comp. Outlet (TP1 A/B)	2	2 (*)	150	WNRF		Driver Weight	: *	lbs
43	Condensate Outlet Drain	2	1 (*)	150	WNRF		Control Panel Weight	: *	lbs
44	Vent	1	1 (*)	150	WNRF		Package Dimension (L x W x H)	: *	in
45	Drain Skid	1	1 (*)	150	WNRF		Control Panel Dimension (L x W x H)	: *	in

46	**INSPECTION AND TESTING**			**ACCESSORIES**
47		Required	Witness	- Inlet Air Filter **(Note 6)**
48	Shop Inspection	●	○	- Silencer
49	Hydrotest	●	○	- Moisture Spearator
50	Leak Test	●	○	- All accessories required for air compressor
51	Function Test	(FAT & SAT)●	●	
52	Compressor & Driver Performance Test	(FAT & SAT)●	●	
53	Material Certification	●	○	
54	Certification and Inspection Records	●	○	

Applicable to	: **PROPOSAL**	**Client**	: **PT. PERTAMINA EP**
Tag No.	: **Z-006 & Z-005**	**Quantity**	: **2 Units**
Service	: **Air Compressor and Air Dryer Package**	**Code & Standard**	: **API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
P & ID No.	: **TMBP-01-ME-SPE-154**	**Manufacturer**	: **As per PML**

AIR DRYER (Z-005)

				Rev.
1	**PROCESS DATA**			
2	**SERVICE DESCRIPTION**	**NORMAL**	**DESIGN**	
3	FLUID HANDLED	**COMPRESSED AIR**		
4	SOURCE	**AIR COMPRESSOR**		
5	INLET AIR OIL CONTENT (ppm)	**(*)**	**(*)**	
6	INLET AIR TEMPERATURE (°F)	**(*)**	**(*)**	
7	INLET AIR RELATIVE HUMIDITY (%)	**(*)**	**(*)**	
8	INLET AIR PRESSURE (psig)	**(*)**	**(*)**	
9	INLET AIR FLOWRATE (scfm)	**(*)**	**(*)**	
10	OUTLET AIR PRESSURE (psig)	**(*)**	**(*)**	
11	OUTLET DRY AIR FLOWRATE (scfm)	**45**	**45**	
12	OUTLET DRY AIR DEW POINT (°F)	**- 40 (at 130 psig)**		
13	OUTLET DRY AIR TEMPERATURE (°F)	**(*)**	**(*)**	
14	PRESSURE DROP **(NOTE 12)** (psi)	**7 (max)**	**7 (max)**	
15	OUTLET DRY AIR OIL CONTENT (ppm)	**(*)**		
16	OUTLET ABSOLUTE FILTER RATING (micron)	**(*)**		
17				
18	AIR DRYER TYPE	**DRY DESICCANT, HEATLESS REGENERATIVE PSA (NOTE14)**		
19	NO. OF DRYER SKIDS	**INSTALLED AT PACKAGED SKID**		
20	NO. OF DRYER VESSELS/ UNIT	**TWO (2) (2 x 100%)**		
21		**(EACH TO BE SIZED FOR MINIMUM 10 MINUTES WITHOUT REGENERATION)**		
22	REGENERATION AIR scfm	**Note 11**	**Note 11**	
23	**INSTRUMENTATION AND CONTROLS (Note 13)**			
24	**REQUIRED INSTRUMENTATION FUNCTIONS:**			

	FUNCTIONS	**LOCAL INDICATION**	**LCP ALARM**	**LCP TRIP**	**DCS**
25					
26					
27	PRE-FILTER HIGH DIFFERENTIAL PRESSURE	√			
28	AFTER FILTER HIGH DIFFERENTIAL PRESSURE	√			
29	DESICCANT VESSEL FLOW				
30	DESICCANT VESSEL PRESSURE	√			
31	DESICCANT VESSEL TEMPERATURE				
32	DEW POINT / MOISTURE CONTENT AT DOWNSTREAM AFTER FILTER	√			√
33	DESICCANT VESSEL A OR B REGENERATING SEQUENCE	√			√
34	POWER ON/ OFF	√	√	√	√
35	FAILURE TO SWITCH	√			
36	COMMON ALARM / FAULT	√	√	√	√
37	HIGH HUMIDITY LEVEL				
38	RUNNING STATUS	√			√
38					
39	**CONTROL PANEL**				

40	TYPE : **Push Button**	SEQUENTIAL CONTROL REQUIRED	: **YES, IP 65 (outdoor with shelter)**
41	MANUFACTURER : **INGERSOLL RAND**	DIMENSIONS (L x W x H) (mm)	: *
42	LOCATION : **LOCAL MOUNTED PANEL**	FACTORY TESTING	: **FUNCTIONAL TEST (WITNESS)**
43			
44			
45			
46			
47			
48			
49			

Applicable to	:	**PROPOSAL**	**Client**	:	**PT. Pertamina EP**
Tag.No.	:	**Z-006 & Z-005**	**Quantity**	:	**2 Units**
Service	:	**Air Compressor and Air Dryer Package**	**Code & Standard**	:	**API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
PID No.	:	**TMBP-01-ME-SPE-154**	**Manufacturer**	:	**As per PML**

INSTRUMENT AIR DRYER SKID (Cont'd)

1	**EQUIPMENTS**		**PRE-FILTERS**	**AFTER FILTERS**	**Rev.**
2			**DESIGN DATA**		
3	VENDOR		(*)	(*)	
4	SERIAL NO		(*)	(*)	
5	TAG NO.		(*)	(*)	
6	QUANTITY		**2 (1 Working + 1 Standby)**	**2 (1 Working + 1 Standby)**	
7	FILTERED FLUID		**INSTRUMENT AIR**	**INSTRUMENT AIR**	
8	INLET PRESSURE OPERATION	(psig)	(*)	(*)	
9	INLET TEMPERATURE OPERATION	(°F)	(*)	(*)	
10	DESIGN PRESSURE	(psig)	(*)	(*)	
11	DESIGN TEMPERATURE	(°F)	(*)	(*)	
12	CONTAMINANTS		(*)	(*)	
13	CORROSIVE COMPOUNDS		(*)	(*)	
14	FLOW RATE, DESIGN	(scfm)	(*)	(*)	
15	MAX ALLOWABLE DP CLEAN	(psi)	(*)	(*)	
16	MAX ALLOWABLE DP DIRTY	(psi)	(*)	(*)	
17	MAX OIL CONTENT (OUTLET)	(ppm)	(*)	(*)	
18	FILTRATION	(micron)	**0.01**	**0.01**	
19	EFFICIENCY	(%)	**99**	**99**	
20	GAS COMPOSITION (MOL%)		**AIR & WATER VAPOUR**	**AIR & WATER VAPOUR**	
21	RELATIVE HUMIDITY (MAX)	(%)	**88%**	**88%**	
22	CORROSION ALLOWANCE	(in)	(*)	(*)	
23	INTERNAL DIA. TOP/BOTTOM	(in)	(*)	(*)	
24	SHELL THICKNESS	(in)	(*)	(*)	
25	COVER THICKNESS	(in)	(*)	(*)	
26	TOTAL VOLUME	(in^3)	(*)	(*)	
27	OPERATING VOLUME	(in^3)	(*)	(*)	
28	CODE		(*)	(*)	
29			**CONSTRUCTION**		
30	ELEMENT TYPE		**CARTRIDGE**	**CARTRIDGE**	
31	MATERIAL		**Alumunium**	**Alumunium**	
32	SCREEN SIZE	(micron)	(*)	(*)	
33	TOTAL FILTER AREA	(in^2)	(*)	(*)	
34	CLEANING		(*)	(*)	
35	INITIATED BY....@ DIFF. PRESSURE		(*)	(*)	
36	VENT VALVE		(*)	(*)	
37	DRAIN TRAP		**AUTO**	**AUTO**	
38	DRAIN VALVE		**AUTOMATIC LIQUID DRAIN TRAP**	**AUTOMATIC LIQUID DRAIN TRAP**	
39	DIMENSIONS	(ft-in)	(*)	(*)	
40	WEIGHTS (DRY/ COVER/ ELEMENT)	(lb)	(*)	(*)	
41	LENGTH CLOSED/OPEN	(ft-in)	(*)	(*)	
42	WIDTH CLOSED/OPEN	(ft-in)	(*)	(*)	
43	HEIGHT INCL. SUPPORT CLOSED/OPEN	(ft-in)	(*)	(*)	
44	FREE HEIGHT REQD. FOR MAINTENANCE	(ft-in)	(*)	(*)	
45	FREE SPACE REQD. IN ANY DIRECTION	(ft-in)	(*)	(*)	
46	MANUFACTURER DRAWING NO.		(*)	(*)	
47	MOUNTING (HORIZONTAL/ VERTICAL)		(*)	(*)	
48	INSULATION THICKNESS		(*)	(*)	
49	DIFF. PRESSURE		**PDI**	**PDI**	
50	COLOR CHANGE MOISTURE CONTENT INDICATOR		(*)	(*)	
51	PRESSURE RELIEF VALVE		(*)	(*)	
52					
53					
54					
55					
56					
57					
58					
59					
60					
61					
62					
63					
64					

Applicable to	: **PROPOSAL**	**Client**	: **PT. PERTAMINA EP**
Tag No.	: **Z-006 & Z-005**	**Quantity**	: **2 Units**
Service	: **Air Compressor and Air Dryer Package**	**Code & Standard**	: **API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
P & ID No.	: **TMBP-01-ME-SPE-154**	**Manufacturer**	: **As per PML**

INSTRUMENT AIR DRYER SKID (Cont'd)

1	**AIR DRYER VESSELS**				Rev.
2	TAG NO. : **Z-005**		QUANTITY : **2 sets 2 chamber per set**		
3	**DESIGN DATA**		**OPERATING CONDITIONS**		
4	CODE	**ASME VIII Div. 1 (No Stamp)**	CONTENTS	**COMPRESSED AIR**	
5	SPECIFICATIONS	**TMBP-SP001-ME-SPE-0105**	SPECIFIC GRAVITY OF CONTENTS	**1 (WATER)**	
6	STATUTORY AUTHORITY APPROVAL	● YES ○ NO	OPERATING PRESSURE (psig)	**145**	
7	DESIGN PRESSURE, INTERNAL (psig)	*	OPERATING TEMPERATURE (°F)	**(*)**	
8	DESIGN PRESSURE, EXTERNAL (psig)	*	OPERATING VOLUME (ft^3)	**N/A**	
9	DESIGN TEMPERATURE (°F)	*	TOTAL VOLUME (ft^3)	**N/A**	
10	DESIGN MIN. TEMP. AT DESIGN PRESSURE (°F)	*	**MATERIALS OF PRESSURE VESSEL (Note 3)**		
11	INTENDED NUMBER OF OPERATING CYCLES	*	SHELL	**SA 516 Gr. 70 (*)**	
12	CORROSION ALLOWANCE (in)	**0.125**	HEADS	**SA 516 Gr. 70 (*)**	
13	JOINT EFFICIENCY	SHELL: * HEADS: *	NOZZLE NECKS	**(*)**	
14	ORIENTATION	○ HORIZONTAL ● VERTICAL	NOZZLE/BLIND FLANGES	**(*)**	
15	WIND LOADS (OPR / DES) (ft/s)	*	SHELL FLANGES	**(*)**	
16	SEISMIC LOADS (Hor / Ver)	*	REINFORCING PADS	**(*)**	
17	OTHER LOADS	*	WEAR PLATES	**(*)**	
18	LENGTH (TAN-TAN) (ft-in)	*	STIFFENING RINGS	**(*)**	
19	INTERNAL DIAMETER (in)	*	LIFTING LUGS	**(*)**	
20	SHELL THICKNESS (in)	*	CLIPS	**(*)**	
21	HEAD THICKNESS (in)	*	INSULATION SUPPORTS	**(*)**	
22	SUPPORT TYPE	**LUG SUPPORT**	SADDLES/SKIRT/LEGS	**(*)**	
23	INSULATION (in)	○ HOT ○ COLD ○ THK.	BASE PLATES	**(*)**	
24	LADDERS & PLATFORMS **(Note 35)**	○ YES ● NO	BRACKET SUPPORTS	**(*)**	
25	INSULATION SUPPORTS	○ YES ● NO	GASKETS	**(*)**	
26	LIFTING LUGS	● YES ○ NO	EXTERNAL FLANGE BOLTS	**(*)**	
27	EARTHING LUGS	● YES (SS316) ○ NO	EXTERNAL FLANGE NUTS	**(*)**	
28	MANWAY COVER / HANDHOLE	○ HINGED ○ WITH DAVIT	STRUCTURAL BOLT	**(*)**	
29	INSTRUMENTATION	*	INTERNAL BOLTS / NUTS	**(*)**	
30			INTERNALS	**(*)**	
31			NAMEPLATE & BRACKET	**(*)**	
31	**FABRICATION & INSPECTION**		**WEIGHT (Note 4)**		
32	RADIOGRAPHY	**AS PER CODE & SPECIFICATION**	DRY WEIGHT (lb)	**(*)**	
33	MAGNETIC PARTICLE		OPERATING WEIGHT (lb)	**(*)**	
34	DYE PENETRANT		INTERNAL WEIGHT (lb)	**(*)**	
35	ULTRASONIC				
36	HYDROSTATIC TEST				
37	MIGAS CERTIFICATE				
38	PWHT				
39	IMPACT TEST	**N/A**			
40	**NOISE LIMITATIONS - SOUND PRESSURE LEVELS**				
41	ACOUSTIC TREATMENT:	*	NOISE LIMIT dB(A)	**85 @ 3ft**	
42	TYPE	**SILENCER / NOISE ENCLOSURE**	ANTICIPATED dB(A)	**85**	
43	SILENCER SCOPE	**VENDOR**			
44	**AIR DRYER CONSTRUCTIONS**				
45	DRYER TYPE	**DESICCANT**	AIR FLOW (Nm3/h)	*	
46	DESICCANT QUANTITY PER CHAMBER (lb)	*	BLOWER TYPE	**N/A**	
47	REGENERATION CYCLE TIME (min)	*	CAPACITY (Nm3/h)	**N/A**	
48	SWING TYPE	**PRESSURE (HEATLESS)**			

49 **NOZZLE CONNECTION FOR DRYER PACKAGE (Note 2)**

50	MARK	QTY	RATING	FACING	SIZE (In)	SERVICE	MATERIAL	REMARK
51	TP1	1	150#	WNRF	2 (*)	AIR INLET	A 106 B	
52	TP2	1	150#	WNRF	2 (*)	AIR OUTLET	A 106 B	
53	TP3	1	150#	WNRF	(*)	DRAIN	A 106 B	
54	TP4	1	150#	WNRF	(*)	VENT	A 106 B	

55 56 57 58 59 60 61 62 63 64 65 66

Applicable to	:	**PROPOSAL**	**Client**	:	**PT. PERTAMINA EP**
Tag No.	:	**Z-006 & Z-005**	**Quantity**	:	**2 Units**
Service	:	**Air Compressor and Air Dryer Package**	**Code & Standard**	:	**API 619, ANSI/ISA-7.0.01 & ISO 8573-1**
P & ID No.	:	**TMBP-01-ME-SPE-154**	**Manufacturer**	:	**As per PML**

		NOTES	**Rev.**
		(* = VTA (VENDOR to Advise and shall fill detail value/data before purchase order)	
	(1)	Painting & Coating shall comply as per TMBP-SP001-PI-SPE-D109Specification – Piping Painting & Lettering Specification	
	(2)	Tie in Point Nozzle Connection is only preliminary design, detail arrangement, elevation & quantity of Nozzle shall be provided by VENDOR before Purchase Order (PO).	
	(3)	All metallic materials within the package shall be provided with Material Certificate in accordance with ISO 10474 Pressure parts with 3.1 certification	
		and Non-pressure parts with 2.2 certification.	
	(4)	Electric motors with rated capacity 10 HP and above shall be equipped with space heater rated voltage 230 V applied at 120 V / 50 Hz / 1 Ph.	
	(5)	Alarm will initiate Common Alarm (non S/D) and Trip will initiate Common Fault (S/D) alarm.	
		Alarm and Trip signal shall be displayed at local panel with annunciator. Each Compressor shall be equipped with local panel.	
		Enclosure material for control panel shall be suitable for NEMA 4 and painted as per TMBP-SP001-PI-SPE-D109 Piping Painting & Lettering Specification	
	(6)	Compressor Air Inlet filter shall be fitted with replaceable cartridge type.	
	(7)	Air Quality Product shall be in accordance with ISO 8573-1 Class 1.	
	(8)	VENDOR shall check, recalculate & guarantee Vessel strength calculation before Purchase Order (PO).	
	(9)	Enclosure material for Air Compressor shall be CARBON STEEL as minimum.	
	(10)	Air Compressor & Dryer Package shall be designed as per Code, standard & specification refer to Page 10.	
	(11)	15% extra instrument air flow for dryer regeneration is included (vendor to confirm and adjust inlet flow based on requirement of air flow for regeneration).	
	(12)	Pressure drop for total package including pre-filter and after filter.	
	(13)	Air dryer shall be equipped with necessary control for automatic switch over between towers.	
	(14)	VENDOR shall provide certification or supporting document for un-witnessed test and state the maximum allowable temperature for desiccant.	